

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

Group and Individual Assignments Briefs

Group Project Instructions for CA1 and CA2

1. All group projects are to self-formed with a **group size of min 4 – max 6 pax**. The members in the group should be diverse in background (different schools, year of study, etc).
2. Students without groups will be randomly allocated to a group before Week 4.
3. The submission is to be
 - a. Powerpoint slides
 - i. **AI Assessment Scale (AIAS) = 2 (two). AI-Assisted Idea Generation and Structuring**
 - ii. AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work.
 - b. An audio and video presentation embedded with Powerpoint slides in **MP4 video format of not more than 10 minutes**. The MP4 video [file size < 500MB] will be played during the Presentation, and students are expected to answer questions after the end of the presentation.
 - i. **AI Assessment Scale (AIAS) = 1 (one). No AI.**
 - ii. This video and recording is to be completed entirely without AI assistance. Each team should rely solely on their own team members' presentation skills.
4. The steps to create a video in PowerPoint can be found here: <https://youtu.be/D8JV3w4TOVw> . Alternatively, students can use other tools (Microsoft Teams, Zoom, etc) to create the MP4 video.
5. **All members of the team shall take turns and have a speaking role** presenting the PowerPoint slides (video-in-video style) in the MP4 video presentation. Penalties will be imposed on both the team and individuals that do not have a speaking role in the video presentation.
6. Students are expected to **work effectively as a team**. Each team member is expected to do her/his share of work conscientiously, in a timely manner and with a cooperative and positive outlook.

Continuous Assignment 1 [CA1] (25%)

Submission due on 16 Sep 2026, 2359 hours

The objective of this CA1 is to be completed by a group of students (min 4 – max 6 pax):

- Examine an existing and successful Technology-Based Innovation against the **Cyclic Innovation Model**.
- Explain how Entrepreneurship brought Scientific Exploration, Technological Research, Product Development and Market Transitions together.
- The following technologies **should not** be considered: Artemisinin, Drones, Water from Reverse Osmosis, Green Steel, Dental Braces, Blue Led, Plant Based Meat, Lab-Grown Meat, Vending Machine, Meals Ready to Eat, Vertical Farming, 3D Printers, Vacuum Cleaners, Ozempic, Nvidia GPU, Progressive Lenses and Life Straw.

The target audience is the Board of Directors of a scientific company who wants to understand how previous Technology-based companies have managed to bridge the gaps between scientists, engineers and marketers in a successful market launch.

The submission will be evaluated on the following rubrics:

Continuous Assignment 1 (CA1) 25% : 20% (Group); 5% (Individual)

Group Component: 20%			
Criteria	Exceeded Expectations 4 % - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Introduction to the existing and successful Technology-Based Innovation and its commercial success. (5%)	Detailed explanation of an existing and successful Technology-Based Innovation and its commercial success.	There is explanation of an existing and successful Technology-Based Innovation and its commercial success.	There is explanation of an existing and successful Technology-Based Innovation and there is lack of discussion on its commercial success.
Criteria	Exceeded Expectations 8% - 10%	Met Expectations 5% - 7%	Below Expectations 0% - 4%
Application of the Cyclic Innovation Model. (10%)	There are clear and detailed findings of - Scientific Exploration - Technological Research - Product Development - Market Transition And strong demonstration on how Entrepreneurship links all these elements.	There are findings of - Scientific Exploration - Technological Research - Product Development - Market Transition And some demonstration on how Entrepreneurship links up these elements.	There is a lack of findings of - Scientific Exploration - Technological Research - Product Development - Market Transition And little or no demonstration on how Entrepreneurship links up these elements.
Criteria	Exceeded Expectations 4 % - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Video presentation of the Powerpoint slides with Embedded Individual Presentation. (5%)	The video presentation (within 10 mins) has clear and concise Powerpoint slides (within 10 mins) that is accompanied by video presentation by all team members that are consistent in tone and audibility.	The video presentation (within or exceeded 10 mins) has Powerpoint slides that is accompanied by video presentation (with unclear audio) and all team members are present.	The video presentation may be incomplete and/or overly lengthy (beyond 12 mins) and/or not presented by all team members.
Individual Component: 5%			
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Performance & Delivery (5%)	The individual video presentation in the embedded is clear & logical.	The individual video presentation is included in the submission.	The individual video presentation is not included.

Continuous Assignment 2 [CA2] (25%)

Submission due on 4 Nov 2026, 2359 hours

The objective of this CA2 is to be completed by a group of students (min 4 – max 6 pax):

- Formulate a New Service that is entrepreneurial and fulfilling market needs.
- The new service theme to be formulated is - **“To make Youth Hub *Scape Cool”**. Refer to this article: <https://www.straitstimes.com/life/is-youth-hub-scape-still-cool>
- This service is to be run during last 3 weeks of December 2026 and from mid-May to end of June 2027 to coincide with the school holidays of polytechnic, secondary and primary school students.
- *“This could be a foodscape or landscape or escape. Whatever you want it to be.”*
- Appraise the possible industry collaboration for market entry.
- Examine the long-term goal in terms of the management of intellectual property and organisational knowledge.

The target audience is the Board of Directors of SCAPE Co. Ltd looking to launch a new service during the December vacation to meet market needs.

The submission will be evaluated on the following rubrics:

Continuous Assignment 2 (CA2) 25% : 20% (Group); 5% (Individual)			
Group Component: 20%			
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Introduction to proposed New Service that is entrepreneurial and fulfils market needs. (5%)	Detailed explanation of a New Service that is both entrepreneurial and fulfils market needs.	There is explanation of a New Service that is both entrepreneurial and fulfils market needs.	There is explanation on the New Service but lacks the discussion on how it is entrepreneurial and/or how it fulfils market needs.
Criteria	Exceeded Expectations 8% - 10%	Met Expectations 5% - 7%	Below Expectations 0% - 4%
Analysis of the possible industry collaboration, intellectual property management and organization knowledge management. (10%)	There are clear and detailed discussions on the possible • Industry Collaboration • Intellectual Property Management • Organization Knowledge Management And strong demonstration on how it will lead to Entrepreneurial value-creation.	There are discussions on the possible • Industry Collaboration • Intellectual Property Management • Organization Knowledge Management And some demonstration on how it will lead to Entrepreneurial value-creation.	There is a lack of discussions on the possible • Industry Collaboration • Intellectual Property Management • Organization Knowledge Management And little/no demonstration on how it will lead to Entrepreneurial value-creation.
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%

Video presentation of the Powerpoint slides with Embedded Individual Presentation. (5%)	The video presentation (within 10 mins) has clear and concise Powerpoint slides (within 10 mins) that is accompanied by video presentation by all team members that are consistent in tone and audibility.	The video presentation (within or exceeded 10 mins) has Powerpoint slides that is accompanied by video presentation (with unclear audio) and all team members are present.	The video presentation may be incomplete and/or overly lengthy (beyond 12 mins) and/or not presented by all team members.
Individual Component: 5%			
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Performance & Delivery (5%)	The individual video presentation in the embedded is clear & logical.	The individual video presentation is included in the submission.	The individual video presentation is not included.

Individual Assignments

1. Students are to use the front cover for the submission of the Individual Assignment 1 & 2.
2. The students are also to make a declaration that the individual assignments completed are their own work.

Individual Assignment 1 [IA1] (20%)

Submission due 27 Sep 2026, 2359 hours

The objective of this IA1 is to for the students:

- Explain the Current Market Needs of an innovative physical product a Singaporean Young Senior (born in 1960-1973). Read about the characteristic of Young Seniors here: <https://www.channelnewsasia.com/commentary/singapore-majulah-package-pioneer-merdeka-generation-support-shift-ndr-2023-3715821>
- Formulate a New Innovative Product with features to meet this Market Need.
- This would include description and sketch(es) of the proposed New Product.

Ideas generation and structuring on the current market needs and formulating a new product features are:
AI Assessment Scale (AIAS) = 2 (two). AI-Assisted Idea Generation and Structuring

This submission is to be completed as an Infographic in **A4 size in Portrait mode**. Infographics is the creative method to provide an easy-to-understand overview through the clever combination of imagery, charts, and text. **It should be completely hand drawn and written**. This is to mimic the scenario of meeting a potential partner/investor at a networking event and wanting to pitch your idea using pen, pencil, paper, napkin is available.

AI Assessment Scale (AIAS) = 1 (one). No AI.

This infographic is to be done using hand drawings and handwriting. IA1 will be evaluated based on the following rubrics:

Individual Assignment 1 (IA1) 20%			
Criteria	Exceeds Expectations 4 % - 5%	Meets Expectations 2% - 3%	Below Expectations 0% - 1%
Introduction to the Current Market Needs (5%)	Detailed explanation of Market Needs using Maslow Hierarchy of Needs.	There is evidence of Market Needs using Maslow Hierarchy of Needs.	Lacking or no evidence of Market Needs using Maslow Hierarchy of Needs.
New Innovative Product Features (5%)	Detailed explanation of How the New Innovative Product features will fulfil the Market Needs.	There are some explanation of how the New Innovative Product features will fulfil the Market Needs.	Brief explanation of how the New Innovative Product features will fulfil the Market Needs.
New Product Sketches (5%)	Detailed sketches of 2 or more views with the key features of the New Product.	Sketches of some of the key features of the New Product.	Sketch(es) of the New Product lacking in details of key features.
Clarity of Infographics (5%)	Clear, concise and well organized infographics with	Contains organized infographics of the New Product	Infographics that are layout in a haphazard manner on the New

	good balance of New Product Description, sketches and the excellent use of colours.	description, sketches and good use of colours.	Product description, sketch(es) and the lack of use of colours.
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Individual Assignment 2 [IA2] (20%)

Submission is due on 15 Nov 2026, 2359 hours

The objective of this IA1 is to for the student to reflect on this module at its conclusion. They are to:

- Provide the reasons for taking up the Module
- Analyze the **three key areas** from this course that have impacted you in your future venture creation as an Entrepreneur and/or Corporate Entrepreneur. The three key areas can include learnt concepts (e.g. Cyclic Innovation Model, Value Creation, IP Management, Organizational Knowledge, etc), and skillsets (e.g. Sketching, Teamwork in New Product/Service Development, etc.) from this course.
- You will need to embed **artifacts for each of the three areas** that you choose to discuss. Artifacts can include any of the following:
 - **Photographs** (e.g. images of your attempts at sketches during the Sketching session, etc)
 - **Screenshots** (e.g. of your research, or speech preparation for the Cyclic Innovation Model presentation or notes you had taken during the lecture session, etc)
 - **Audio recording of up to 1 min** (e.g. about your own immediate thoughts after hearing from the guest speaker, etc)
 - **Video recording of up to 30 seconds** (e.g. Team and teammates working together on the group project, etc)
 - Or, any other relevant artifacts.
- Conclude on how this course will help in your own future entrepreneurial endeavors.

AI Assessment Scale (AIAS) = 2 (two). AI-Assisted Idea Generation and Structuring

Each student should reflect solely on their own learnings from the knowledge, understanding and skills from the course and include artifacts, such as photographs, screenshots, audio recordings, video recordings, etc.

This submission is to be evaluated based on the following rubrics:

Individual Assignment 2 (IA2)			
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Introduction to the Reason for taking up this Module. (5%)	Detailed explanation on the reasons for taking up this Module.	There are some explanation on the reasons for taking up this Module.	Lacking Introduction and/or no explanation on the reasons for taking up this Module.
Criteria	Exceeded Expectations 8% - 10%	Met Expectations 5% - 7%	Below Expectations 0% - 4%
Analyze the concepts, tools and skills learn in this module that are relevant to future venture creation. (10%)	There is clear and detailed analysis of • Concepts (Value Creation, IP Management,	There is some analysis of • Concepts (Value Creation, IP Management,	There is a lack of analysis of • Concepts (Value Creation, IP Management,

	Organizational Knowledge) Skillset (Sketching, Teamwork in New Product/Service Development, etc) And strong demonstration on how these are relevant to their future venture creation.	Organizational Knowledge) Skillset (Sketching, Teamwork in New Product/Service Development, etc) And demonstration on how these are relevant to their future venture creation.	Organizational Knowledge) Skillset (Sketching, Teamwork in New Product/Service Development, etc) And lacks demonstration on how these are relevant to their future venture creation.
Criteria	Exceeded Expectations 4% - 5%	Met Expectations 2% - 3%	Below Expectations 0% - 1%
Conclusion (5%)	Detailed conclusion on how the Module will help in the entrepreneurial journey.	There are some conclusion on how the Module will help in their entrepreneurial journey.	Lacking Conclusion and/or no explanation on how the Module will help in their entrepreneurial journey.